

Heart Attack Detection with Machine Learning: A Multi-Feature Classification Study

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1 Introduction

1.1 Background

Every 34 seconds, an American has a heart attack, with over 800,000 cases recorded annually in the United States. This occurs when plaque buildup blocks blood flow to the heart and is the culmination of long-term physiological deterioration driven by both modifiable and non-modifiable risk factors—including but not limited to hypertension, obesity, smoking, diabetes, and also factors such as age, sex, and family history. While technological and medical advancements have drastically improved survival rates, heart disease remains the leading cause of death in the US [1] [2][3][4]. Critically, a significant proportion of heart attacks occur in individuals who were never previously identified as high risk, as many have underlying silent or atypical symptoms. This highlights a diagnostic gap that underscores the need for more comprehensive, data-driven approaches to accurately determining whether an individual has experienced a cardiac event.

1.2 Motivation

Cardiovascular disease disproportionately affects individuals across a wide range of demographic and health profiles, yet traditional clinical screening tools often rely on a narrow set of biomarkers. With cardiovascular diseases (heart disease) being the leading cause of death globally and in the United States, heart attacks are also a major complication of them. Machine learning offers a complementary approach—one that is capable of surfacing complex, non-linear relationships between lifestyle factors, chronic conditions, cardiac outcomes, and other health metrics. By exploring factors and using a broad set of health indicators drawn from self-reported survey data, including pre-existing conditions

such as diabetes and asthma, mental and general well-being, and other behavioral and physiological features, this study seeks to develop a classification model that determines whether an individual has or has not experienced a heart attack. By leveraging a data-driven methodology, the goal is to produce a scalable and interpretable tool that can accurately distinguish between these two outcomes across diverse patient profiles.

2 Exploratory Data Analysis

2.1 Dataset Description

Our dataset, sourced from Kaggle, originates from the CDC's Behavioral Risk Factor Surveillance System (BRFSS), which conducts annual telephone surveys to collect health status data from U.S. residents across all 50 states, the District of Columbia, and three U.S. territories. The 2022 survey cycle forms the basis of this dataset, encompassing over 400,000 adult respondents. The original BRFSS survey contains nearly 300 variables, which were reduced to 40 relevant health indicators for this dataset, covering factors such as diabetes status, BMI, smoking history, physical activity, alcohol consumption, mental well-being, and various chronic conditions. Our target variable is HadHeartAttack, a binary indicator of whether a respondent has ever been told by a healthcare provider that they experienced a heart attack.

2.2 Data Cleaning Process

2.2.1 Data Cleaning Process. The dataset consisted of features with data types float64 and object. Numerical features (float64) were retained without modification. Features stored as object were treated as categorical variables and processed according to their type. Ordinal categorical features were encoded numerically

and converted to float64 when appropriate. For ordinal categories corresponding to numerical ranges, the midpoint of each range was used instead of arbitrary integer encodings (e.g., 1, 2, 3, 4), allowing the encoded values to better reflect the underlying quantitative relationship. Nominal categorical features were preserved as categorical strings to enable one-hot encoding in models that require numerical inputs while remaining compatible with models that can directly handle categorical data.

2.2.2 Handling Missing Values. Missing values in all numerical features, including encoded ordinal variables, were imputed using the median, which is robust to outliers and skewed distributions. Additionally, a binary missingness indicator was created for each imputed feature to capture potential predictive information contained in the missingness pattern itself. For nominal categorical features, missing values were not imputed. Instead, a new category labeled “Missing” was introduced to explicitly represent missing observations while preserving information about their occurrence.

2.3 Findings from EDA

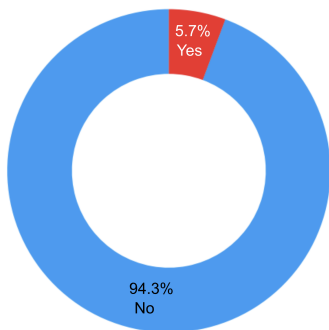


Figure 1: Distribution of HadHeartAttack (Target Variable)

2.3.1 Basic Statistics. The distribution of the target variable, HadHeartAttack, is highly imbalanced: 94.3% of individuals did not report having a heart attack, while only 5.7% did. Because of this class imbalance, accuracy alone would be a misleading measure of model performance. Therefore, evaluation metrics that account for both false positives and false negatives, such as the F1 score or ROC-AUC, will be used to assess model

performance. Among the continuous features, HeightInMeters and SleepHours are approximately normally distributed. BMI and WeightInKilograms exhibit moderate right skewness, while PhysicalHealthDays and MentalHealthDays are strongly right-skewed, with most individuals reporting relatively few unhealthy days and a smaller number reporting many. For the binary health condition indicators, such as HadAngina and HadKidneyDisease, the vast majority of observations belong to the negative class, with relatively few positive cases. This distribution is consistent with expected population-level prevalence rates for these conditions.

2.3.2 Properties.

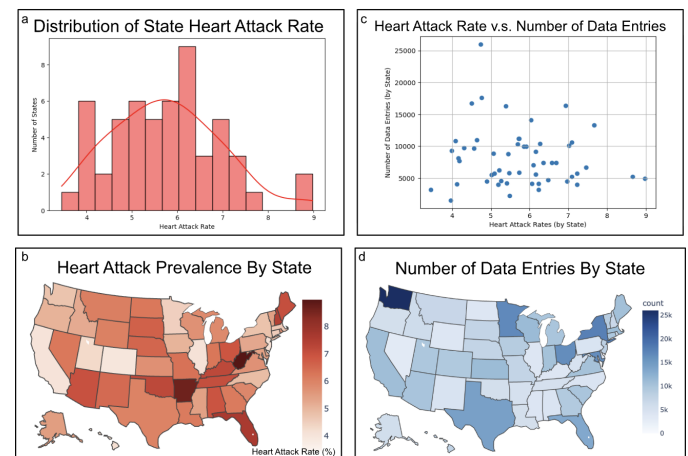


Figure 2: Inspecting the State Feature

The distribution of heart attack rates by states is roughly normal with slight right skew (fig. 2a). The choropleth of Heart Attack Rates (fig. 2b), shows Arkansas and West Virginia have noticeably higher rates of heart attack, which is consistent with the right skew on the histogram, while DC has lower rates. This suggests that state may be a useful feature for our model. No strong correlation between the two variables (fig. 2c). This suggests that sample size (data entries) does not introduce systematic bias toward higher or lower reported heart attack rates, which supports the validity of comparing states directly. The notably different geographic pattern in fig. 2b versus fig. 2d serves as a further visual confirmation of this.

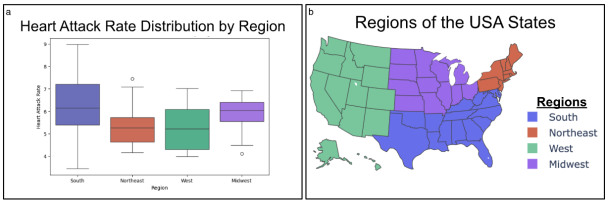


Figure 3: State and Region

In Figure 3, states are grouped into geographic regions (shown in Figure 3b), and heart attack rates are visualized using boxplots that summarize the distribution of state-level rates within each region. The plots suggest that different regions exhibit distinct central tendencies and variability in heart attack rates. Based on these observed regional differences, an additional feature, Region, was created by mapping each state to its corresponding geographic region using the provided state-level data.

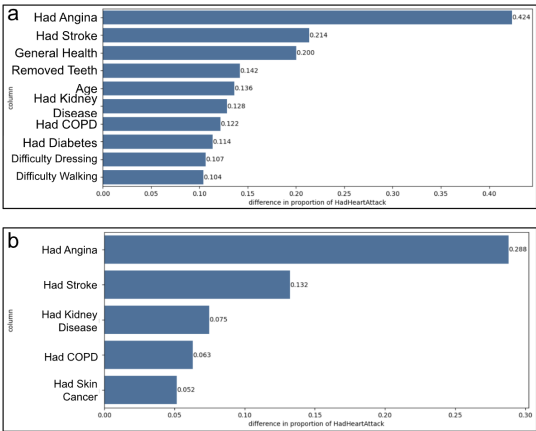


Figure 4: Feature and Feature Missingness Importance

Both graphs in fig. 4 highlight the largest differences in heart attack prevalence across categories within the same feature. Fig. 4a shows categorical variables where the difference in heart attack prevalence across categories exceeds 10 percentage points. Fig. 4b shows missingness indicators, where the prevalence differs by more than 5 percentage points between missing and non-missing groups.

Notably, both HadAngina and the missingness of HadAngina are associated with substantial differences in heart attack prevalence. This suggests that not only

the feature itself, but also its missingness pattern, may carry meaningful predictive information.

2.3.3 Interesting Findings.

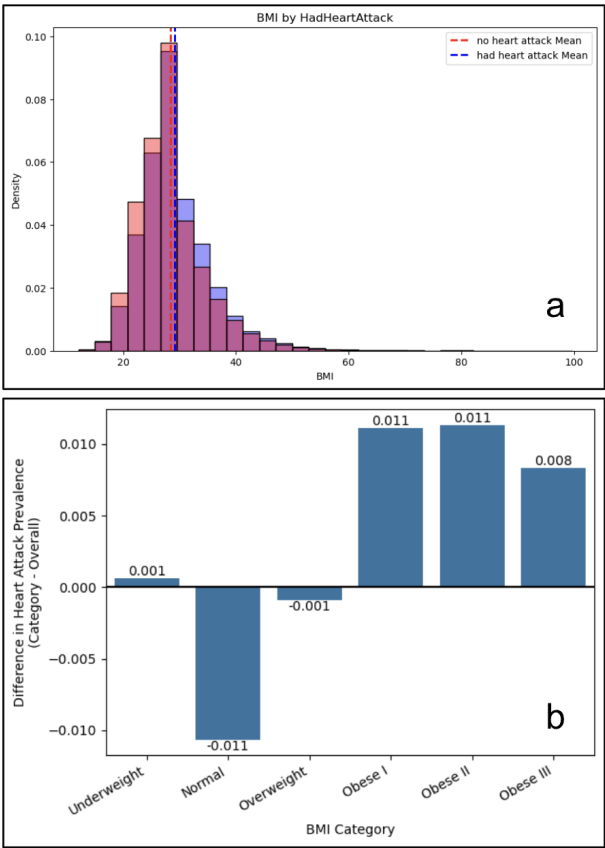


Figure 5: BMI and Heart Attack

Looking at the fig 5a, BMI appears to have only a modest relationship with heart attack status, as the distributions for individuals with and without heart attacks show similar shapes and central tendencies.

However, examining heart attack prevalence across BMI categories (fig 5b) reveals a more nuanced relationship. Prevalence varies across BMI ranges, with slightly elevated rates observed within the underweight category and very elevated rates observed within the obese categories. This pattern suggests a non-linear relationship between BMI and heart attack risk, where both extremes of BMI are associated with increased risk.

3 Predictive Task

3.1 Identify Predictive Task

This study aims to classify whether an individual has experienced a heart attack by leveraging a comprehensive set of personal health indicators and established cardiovascular risk factors. To achieve this, we aim to develop a classification model trained on an extensive dataset comprising over 400,000 observations from across individuals in the United States. In constructing our final predictive model, we will evaluate its performance against 4 baseline models—Logistic Regression, CatBoost, Random Forest Classifier, and Naive Bayes Classifier—and iteratively improve upon them to create a more robust and accurate model.

3.2 Evaluation of Different Models

The primary metric used to compare our baseline and final models is the F1-score for the positive case (has had a heart attack). The F1-score is the harmonic mean of precision and recall, making it particularly well-suited for imbalanced classification tasks where accuracy alone can be misleading. Unlike accuracy, the F1-score accounts for both false positives and false negatives, providing a more honest measure of model performance when one class heavily outnumbers the other. This fits the case in heart attack classification, where a model could achieve deceptively high accuracy by simply predicting the majority class [6]. We focus on performance for the positive class because accurately identifying individuals who have experienced a heart attack is critical. Additionally, the dataset exhibits class imbalance, with heart attack cases representing the minority class, making evaluation of positive-case predictions particularly important.

3.3 Baseline Models

The baseline model we chose was logistic regression. We thought this was appropriate because logistic regression is a natural baseline for binary classification given its simplicity, interpretability, and historical prevalence in previous models for other binary health outcomes. It predicts categorical binary outcomes (ie. yes/no), which is exactly what we are trying to predict. The coefficients of logistic regression directly quantify each feature's contribution to the prediction, making results easy to interpret and compare. However, logistic regression

assumes a linear decision boundary between classes, which may fail to capture more complex, non-linear relationships between health indicators that likely influence whether an individual has experienced a heart attack.

Following logistic regression, we explored CatBoost, which is a gradient boosting algorithm that builds trees sequentially, with each tree correcting the errors of the previous one. CatBoost is particularly well-suited for health classification tasks because it handles categorical features natively without requiring manual encoding, reducing the risk of information loss during preprocessing. As a boosting method, it places greater emphasis on misclassified instances at each iteration, allowing it to better capture difficult or borderline cases that ensemble methods like Random Forest may overlook. CatBoost also incorporates built-in regularization techniques that help prevent overfitting, and is known to perform well with minimal hyperparameter tuning. This combination makes CatBoost a compelling second model choice. However, a key drawback of CatBoost is that, as a sequential boosting method, it will be more computationally expensive and slower to train, making it less practical for large-scale or real-time applications.

In addition to Catboost, we also explored a Random Forest Classifier to address its core limitations. Since Random Forest is an ensemble method that aggregates predictions across multiple decision trees, it is able to naturally capture more complex, non-linear relationships and feature interactions that our logistic regression's linear decision boundary cannot. This is especially relevant for heart attack prediction, where risk factors like age, blood pressure, cholesterol, and lifestyle habits rarely operate independently but interact in nuanced ways. Random Forest also provides built-in feature importance scores, preserving the interpretability we valued in logistic regression while delivering greater predictive power, and its bagging approach makes it more resistant to overfitting and better suited for generalizing to unseen patients.

3.4 Baseline Results

When interpreting these results, precision refers to the accuracy of positive predictions (out of all the instances the model said were positive, how many were actually

Table 1: Test Set Results — Block 1 (Logistic Regression, Numeric Features Only)

Class	Prec.	Rec.	F1	Support
0.0 (No Heart Attack)	0.98	0.86	0.92	41,696
1.0 (Heart Attack)	0.23	0.70	0.35	2,511
Accuracy			0.85	44,207
Macro Avg	0.60	0.78	0.63	44,207
Weighted Avg	0.94	0.85	0.88	44,207
ROC-AUC			0.8604	

correct), and recall measures completeness (out of all the truly positive things, how many did we manage to catch). F1-Score is the harmonic mean of both precision and recall, and support refers to the number of actual occurrences (ground truth instances) of each class in our specified dataset. Our baseline model achieves an F1-score of 0.92 for the negative class (no heart attack) and 0.35 for the positive class (heart attack), indicating strong performance on the majority class but limited ability to reliably identify individuals who have had a heart attack due to class imbalance.

Another metric is ROC AUC, which stands for Receiver Operating Characteristic Area Under the Curve. It is a standard performance metric used in machine learning and statistics to evaluate how well a classification model can distinguish between different classes. In this case, our ROC AUC is approximately 86%, which is considered clinically useful for a heart attack risk predictor. It indicates that the model has a good ability to differentiate between patients who have had a heart attack and those who have not. A value of 1.0 indicates perfect discrimination. AUC values above 0.80 are generally considered clinically useful, while values below 0.80 are considered of limited clinical utility [7].

Table 2: Test Set Results — Naive Bayes

Class	Prec.	Rec.	F1	Support
0.0 (No Heart Attack)	0.98	0.83	0.90	83,392
1.0 (Heart Attack)	0.19	0.65	0.29	5,022
Accuracy			0.82	88,414
Macro Avg	0.58	0.74	0.59	88,414
Weighted Avg	0.93	0.82	0.86	88,414
ROC-AUC			0.8122	

Naive Bayes achieves an ROC-AUC of 0.8122, which is lower than the Logistic Regression and CatBoost models. The model obtains a recall of 0.65 for the heart attack class, indicating that it identifies a majority of positive cases. However, the precision for the positive class is only 0.19, meaning that many of its positive predictions are false positives. This results in an F1-score of 0.29 for the minority class, which is lower than both Logistic Regression (0.35) and CatBoost (0.34). While Naive Bayes is computationally efficient and relatively simple to train, its performance suggests that it is less effective at balancing sensitivity and precision for this heart attack prediction task.

Table 3: Test Set Results — CatBoost (with and without Hyperparameter Tuning)

Class	Prec.	Rec.	F1	Support
0.0 (No Heart Attack)	0.98	0.83	0.90	83,392
1.0 (Heart Attack)	0.21	0.78	0.34	5,022
Accuracy			0.82	88,414
Macro Avg	0.60	0.80	0.62	88,414
Weighted Avg	0.94	0.82	0.87	88,414
ROC-AUC			0.8905	

CatBoost achieved the highest ROC-AUC score among the evaluated models (0.8905), indicating strong discriminative ability. However, its F1-score for the heart attack class was 0.34, slightly below the Logistic Regression model’s 0.35, due to low precision on the minority class. The tuned and untuned CatBoost models produced

identical classification reports and ROC-AUC scores. Hyperparameter tuning was performed using 15 parameter combinations generated by ParameterSampler for computational efficiency, suggesting that the default CatBoost configuration was already near-optimal for this dataset.

The two Random Forest classifiers share the same ROC-AUC score of 0.8771, indicating similar overall discriminative ability, but differ substantially in their handling of the minority class. The initial model was trained using default parameters and the standard classification threshold of 0.5. Although it achieved high overall accuracy (95%), it performed poorly on heart attack detection, achieving a recall of only 0.12 and an F1-score of 0.20 for the positive class due to the dataset’s

Table 4: Test Set Results — Random Forest (without Hyperparameter Tuning)

Class	Prec.	Rec.	F1	Support
0.0 (No Heart Attack)	0.95	1.00	0.97	41,696
1.0 (Heart Attack)	0.61	0.12	0.20	2,511
Accuracy			0.95	44,207
Macro Avg	0.78	0.56	0.59	44,207
Weighted Avg	0.93	0.95	0.93	44,207
ROC-AUC			0.8771	

Table 5: Test Set Results — Random Forest (with Hyperparameter Tuning)

Class	Prec.	Rec.	F1	Support
0.0 (No Heart Attack)	0.97	0.96	0.96	41,696
1.0 (Heart Attack)	0.43	0.53	0.47	2,511
Accuracy			0.93	44,207
Macro Avg	0.70	0.74	0.72	44,207
Weighted Avg	0.94	0.93	0.94	44,207
ROC-AUC			0.8771	

class imbalance. To address this issue, the second Random Forest model was evaluated using threshold tuning. Classification thresholds ranging from 0.1 to 0.6 were tested on the validation set, and the threshold that maximized the positive-class F1-score was selected and applied unchanged to the test set. This increased recall from 0.12 to 0.53 and improved the positive-class F1-score from 0.20 to 0.47, demonstrating that threshold tuning can substantially improve performance on imbalanced classification tasks such as heart attack prediction.

4 Model

4.1 Heart Attack Prediction Model

The model we propose for this binary classification task is the Random Forest classifier with threshold tuning. After evaluating four candidate models — Logistic Regression, CatBoost, Naive Bayes, and Random Forest — the Random Forest with an optimized decision threshold achieved the highest F1-score of 0.47 on the positive class (heart attack), outperforming all other models

on this primary metric. Random Forest is an ensemble learning method that builds multiple decision trees and aggregates their predictions, making it well-suited for capturing the complex, non-linear relationships between health indicators that are unlikely to be represented by a linear boundary. Its native support for class weighting via `class_weight='balanced'` helps address the significant class imbalance present in this dataset, where only 5.7% of respondents reported having had a heart attack. Furthermore, the addition of threshold tuning — selecting the optimal classification threshold on the validation set rather than defaulting to 0.5 — allowed the model to achieve a more meaningful balance between precision (0.43) and recall (0.53), which is critical in a health classification context where both false positives and false negatives carry real consequences. For these reasons, the threshold-tuned Random Forest represents the most effective and well-justified model for this task.

4.2 Feature Preprocessing

Feature preprocessing followed the data cleaning procedure described previously. Ordinal categorical features were numerically encoded, using the midpoint of the corresponding numerical range whenever possible rather than arbitrary integer values. Nominal categorical features were one-hot encoded to enable their use in models that require numerical inputs.

4.3 Model Optimization

4.3.1 Hyperparameter Tuning. For our Random Forest classifier—the highest performing model—the primary hyperparameters were `n_estimators=100`, `class_weight='balanced'`, and an optimized classification threshold. The number of trees was set to 100 to balance the model stability and computational efficiency. Initially, we started off with `n_estimators=10`, which proved to be insufficient and led to poor recall on the positive class. The hyperparameter `class_weight='balanced'` was used to automatically correct for the 94.3%/5.7% class imbalance in the underlying data we were using, respectively, the proportion of people who didn't have heart attacks to the proportion of people who did. Through hyperparameter tuning, the most sensitive hyperparameter was the classification threshold, tuned by sweeping values from 0.10 to 0.55 and selecting the value that maximized F1-score on the validation set—the default

threshold of 0.5 produced an F1 of 0.20, while the optimized threshold improved this to 0.47, confirming that threshold selection is the most critical design decision in this pipeline.

5 Results

5.1 Comparing the Models

Our proposed Random Forest classifier with threshold tuning outperforms all baseline models on the primary metric of F1-score for the positive class (heart attack). We are primarily using the metric of F1-score for the positive class because our dataset is heavily imbalanced, with only 5.7% of respondents having reported having had a heart attack. This means the overall accuracy alone could be misleading, as a model would achieve a 94% accuracy by simply predicting no heart attack every time. However, the F1-score accounts for both false positives and false negatives, making it a more honest and robust measure of how well the model identifies the minority class that matters most in this classification task.

Logistic Regression achieved an F1 of 0.24, Naive Bayes 0.29, and CatBoost 0.34, while Random Forest achieved 0.47 — a significant improvement over the logistic regression baseline and a meaningful gain over CatBoost. The performance gap is most pronounced in precision, where Random Forest (0.43) substantially outperforms all other models, while maintaining competitive recall (0.53).

Random Forest outperforms the baselines for several reasons. First, as an ensemble of decision trees, it naturally captures non-linear relationships and feature interactions between health indicators that logistic regression's linear decision boundary cannot model. Second, the inclusion of full preprocessing — one-hot encoding categorical features such as State, Sex, and RaceEthnicityCategory alongside standard scaling of continuous features — provided the model with a richer and more informative feature representation. Third, threshold tuning was the single most impactful design decision, nearly doubling F1 from 0.20 to 0.47 compared to the default threshold.

However, the RF Classifier model is not without limitations. A recall of 0.53 means the model still misses

nearly half of all true heart attack cases, which remains a concern in a health context. CatBoost achieved higher recall (0.78) at the cost of much lower precision (0.21), suggesting there is still room to optimize the precision-recall tradeoff further. Future work could explore more exhaustive threshold sweeps, additional feature engineering, or models such as XGBoost or neural networks to close this gap.

5.2 Major Takeaways

5.2.1 Ablation Study. To evaluate the contribution of key design decisions in our final model, we conducted an ablation study by systematically removing or degrading individual components and observing the impact on F1-score for the positive class (heart attack). The components examined were: preprocessing quality, threshold tuning, and feature set composition.

Preprocessing (OHE + Scaling): As demonstrated in our two baseline logistic regression models, we performed full preprocessing on one and restricted the model to numeric-only features in the other, which caused the F1-score to drop from 0.34 to 0.24 on the positive class. This confirms that proper encoding of categorical features such as State, Sex, and RaceEthnicityCategory contributes meaningfully to model performance.

Threshold Tuning: Removing threshold tuning and reverting to the default 0.5 decision boundary caused the Random Forest F1-score on the positive class to drop from 0.47 to 0.20, while recall fell from 0.54 to 0.12. This is the single most impactful design decision in our pipeline. Without it, the model defaults to predicting the majority class the vast majority of the time, rendering it ineffective for this task.

Class Weighting: Removing `class_weight='balanced'` from the Random Forest would further bias the model toward the majority class, compounding the effect of the class imbalance already present in the dataset and likely reducing recall on the positive class further.

Overall, the ablation study confirms that threshold tuning and full preprocessing are the two most critical components of our pipeline, and that degrading either results in meaningful performance drops on the metric most relevant to this classification task.

5.3 Case Study

For the Random Forest model, the recall improved substantially from 0.12 to 0.54 following a series of targeted modifications. This improvement is particularly important in the context of heart attack classification, where high recall is critical to ensure that the model correctly identifies the majority of true cardiac events, thereby minimizing the risk of false negatives and patients being incorrectly classified as low-risk.

Both Random Forest models were trained on identical underlying datasets; however, performance differences arose from changes in model configuration and decision thresholds. In particular, threshold tuning and parallelization via `n_jobs = -1` were introduced. The decision threshold was systematically evaluated over the range 0.1 to 0.6 using the validation set, with the optimal threshold selected based on maximization of the F1-score for the positive class. Lowering the classification threshold increased the number of predicted positive cases, thereby improving recall at the expense of reduced precision. Additionally, enabling full CPU parallelization reduced training time, facilitating more efficient experimentation without affecting predictive performance.

5.4 Previous Literature

Heart attack prediction and cardiovascular risk classification have been well-studied areas of machine learning research. The UCI Heart Disease Dataset has served as the most common benchmark, with studies using logistic regression, decision trees, and ensemble methods reporting classification accuracies between 80–87%, with Random Forest and Gradient Boosting consistently among the top performers. More recent work has leveraged the CDC’s Behavioral Risk Factor Surveillance System (BRFSS), a large-scale self-reported health survey, demonstrating that indicators such as general health status, diabetes, BMI, and smoking history carry significant predictive signal for cardiovascular outcomes, with some models achieving accuracies upward of 90%. There are also several cardiac risk calculators avail-

able for public access, which use the personal health information of an individual to calculate a 10-year and lifelong risk of heart disease. The risk score helps inform individuals about lifestyle changes and can recommend

medicines to prevent heart disease [5], helping lower the possibility of individuals getting heart attacks. Our work aligns with this literature in its goal of binary heart attack classification, while contributing a systematic comparison of multiple classification approaches across a broad feature set including chronic conditions, mental well-being, and lifestyle factors to best distinguish individuals who have experienced a heart attack from those who have not.

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A Appendix

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